

Anna Kalygina

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Education

University of Oxford

Oct 2021 - Jun 2025

BSc in Biological Sciences. UK First-class honours.

Oxford, UK

MSc in Computational Biology. Best Research Dissertation in the field of Genome Diversity and Evolution in the final (4th) year of MBiol.

Bioinformatics Institute

Sep 2024 - Jun 2025

1-year [advanced training in bioinformatics](#) with competitive admissions.

Remote

Letovo International School

Sep 2019 - Jun 2021

International Baccalaureate (IBDP): total: 43/45

Moscow, Russia

Russian National State Diploma: Magna Cum Laude Diploma in 11 subjects

School of Molecular and Theoretical Biology

Aug 2019 - Aug 2021

Competitive [summer research experience](#) for high school students.

Barcelona, Spain

Intern in the Comparative Genomics and Data Analysis Lab

Research Experience

ETH Scientific Assistant

Oct 2025 - Present

Researcher at the [Boeva Lab](#)

Zurich, Switzerland

- Virtual Cell with Latent Reasoning: exploring recurrent architectures that maintain a latent reasoning state to predict cellular responses under interventions (CRISPR/drugs/doses).

Master's Thesis

Sep 2024 - Jun 2025

Graduate student in the [Steve Kelly Lab](#)

Oxford, UK

- Engineered a species-agnostic CNN for *de novo* prediction of promoter location and their effect on gene expression based on the physical DNA shape (pyTorch, Bash, AWK).
- Identified functionally constrained structural motifs across species. Assessed regulatory effect of DNA shape on gene activity.

Stanford Research Internship

Jun 2024 - Sep 2024

Research intern in the [Altemose Lab](#)

Stanford, USA

- Developed a full processing and training pipeline for predicting Hi-C chromatin contact maps from single-molecule Fiber-seq data [[FiberFold](#), [GitHub](#)]. (Nextflow, Hydra, Lightning)
- Implemented single-fiber DNA embeddings. This work presents a novel approach to integrating single-molecule sequencing with deep learning to resolve haplotype-specific Mendelian conditions.

Max Planck Institute

Jul 2022 - Sep 2022

Intern in the lab of [Molecular Mechanisms of Transmembrane Trafficking](#)

Goettingen, Germany

- Implemented a sliding window algorithm to analyze particle colocalization on EM images (Python, Nextflow).
- Simulated large protein interaction through writing macros for AlphaFold. Researched recycling protein pathways through cryo-electron microscopy and crystallography.

Industry Experience

Ufonia Research Internship

Jul 2025 - Sep 2025

AI engineer at [Ufonia](#) startup

London, UK

- Implemented an LLM-as-a-judge for automatic and scalable evaluation of generative performance of conversational agents in clinical conditions.
- Fine-tuned open-source LLMs on multilingual corpus of clinical conversations.

Google STEP Internship

Software Engineer in the *Content API Shopping Team*

Jul 2023 - Sep 2023

Zurich, Switzerland

- Automated Quota assignment to merchants as a part of API design (Java).
- Designed configuration files for the use between several microservices (GCL) and performed analysis on the quota usage (SQL).

Awards

Best MBiol Research Project: Awarded to the best Computational Biology final-year dissertation in Genome Diversity and Evolution. (*Out of 107 students cohort*) **2025**

Letovo School Scholarship: Full scholarship covering the 4-year Integrated Master's course at Oxford University. **2021-2025**

Palantir Scholarship for Women in Technology: Awarded for excellence in science and technology, supporting women pursuing advanced studies. **2024**

Oxford Scholars Award: Styring Junior Scholarship for outstanding performance in exams, tutorials, and classes. **2023-2024**

All-Russian National Olympiad in Biology and Ecology: Consecutive prize winner, ranked in the top 1% of 50,000+ participants. **2019-2021**

Publications & Presentations

Dubocanin, D., **Kalygina, A.**, *et al.* "Integrating Single-Molecule Sequencing and Deep Learning to Predict Haplotype-Specific 3D Chromatin Organization in a Mendelian Condition." *bioRxiv*. doi:10.1101/2025.02.26.640261 **2025**

Ellis, Z., Joselowitz, J., Deo, Y., He, Y. V., **Kalygina, A.**, Higham, A., Rahimzadeh, M., Habli, I., Lim, E. "WER is Unaware: Assessing How ASR Errors Distort Clinical Understanding in Patient-Facing Dialogue." *IWSDS 2026. ACL Ontology. Conference Proceedings* (to appear). **2025**

Kalygina A. "Characterizing structural landscape of core eukaryotic promoters reveals high extent of shape conservation." *Oxford Biomedical Research Centre Conference. Poster Presentation*. **2024**

Skills

Programming languages: Python, R, Bash, Java, GCL, LaTeX, SQL

Tools: Terminal, Git and GitHub, Jupyter Notebook, RStudio, pyBigWig, HiC-Pro, PyMOL, Chimera, ImageJ, SLURM, ML packages (pyTorch, keras, SciPy, TensorFlow)

Wet Lab skills: Work with prokaryotic cultures, bacterial transformation and protein synthesis, construct design, Golden Gate cloning, PCR, nanopore sequencing, Western Blot, PAGE electrophoresis.